

Retail Sales Analysis Dashboard using Power BI

1. Project Overview

Objective

The objective of this project is to analyze retail sales data and identify trends, profitability issues, and business opportunities using Power Query and Power BI. The project focuses on transforming raw sales data into meaningful insights that can support business decision-making.

Tools Used

- Power Query
 - Power BI
 - DAX (Data Analysis Expressions)
 - Microsoft Excel
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2. Dataset Description

The dataset contains retail sales transaction records including:

- Row ID
- Order ID
- Order Date
- Ship Date
- Ship Mode
- Segment
- Category
- Sub-Category
- Sales
- Quantity
- Profit
- Region
- Postal Code

The data was used to evaluate sales performance, profitability, customer segments, shipping methods, and regional trends.

3. Data Cleaning Process

The dataset was cleaned using Power Query following a structured approach.

Understanding Dataset

- Reviewed all columns and understood the business purpose of the dataset.
- Identified the dataset as a retail sales dataset.

Data Type Validation

- Converted Order Date to Date format.
- Converted Ship Date to Date format.
- Corrected Postal Code data type.

Data Standardization

- Standardized text values using proper capitalization for consistency.

Business Rules Validation

- Verified that Order Date is always before or equal to Ship Date.
- Confirmed that same-day shipping records are valid.

Uniqueness Check

- Grouped Row ID and validated that each Row ID is unique.

Category Validation

- Verified that each Sub-Category belongs to the correct Category.

Outlier Validation

- Checked that Sales values are not negative.
- Checked that Quantity values are not negative.
- Verified Discount values are within acceptable limits.
- Reviewed Profit values and found no extreme outliers.

Missing Values Check

- Confirmed that no blank values exist in the dataset.

Duplicate Records Check

- Confirmed that no duplicate records exist.
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4. KPI Measures Created

The following DAX measures were created:

Total Sales

Represents the total revenue generated from all sales transactions.

Total Profit

Represents the total profit earned from all transactions.

Total Quantity

Represents the total number of products sold.

Profit Margin

Calculated as:

Profit Margin = Total Profit / Total Sales

5. Dashboard Development

An interactive Power BI dashboard was created to visualize business performance.

KPI Cards

- Total Sales
- Total Profit
- Total Quantity
- Profit Margin

Visualizations

- Sales by Region
- Profit by Region
- Sales by Category
- Sales by Year
- Sales and Profit Tables

Interactive Slicers

- Region
- Segment
- Category

- Ship Mode

These slicers allow users to perform detailed analysis and drill down into specific business areas.

6. Key Insights

Regional Performance

Region	Profit	Quantity	Sales	Margin
Central	39.87K	9K	503.17K	0.08
South	46.75K	6K	391.72K	0.12
East	94.88K	11K	691.83K	0.14
West	110.80K	12K	739.81K	0.15

Insights:

- West region generated the highest sales, profit, and profit margin.
 - Central region generated the lowest profit and lowest margin despite having the second-highest sales.
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Ship Mode Performance

Ship Mode	Profit	Quantity	Sales	Margin
First Class	49.01K	6K	351.75K	0.14
Second Class	58.96K	8K	466.67K	0.13
Same Day	16.16K	2K	129.27K	0.13

Insights:

- Same Day shipping generated the lowest sales and profit.
 - Standard Class generated the highest sales and profit.
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Category Performance

Category	Profit	Quantity	Sales	Margin
Furniture	19.73K	8K	754.75K	0.03

Category	Profit	Quantity	Sales	Margin
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Office Supplies	126.02K	23K	731.89K	0.17
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Technology	146.54K	7K	839.89K	0.17
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Insights:

- Technology generated the highest profit.
- Furniture generated very low profit despite generating strong sales.
- Furniture has the lowest profit margin among all categories.

Segment Performance

Segment	Profit	Quantity	Sales	Margin
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Consumer	136.37K	20K	1.17M	0.12
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Corporate	94.25K	12K	715.81K	0.13
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Home Office	61.68K	7K	440.07K	0.14
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Insights:

- Consumer segment generated the highest sales and profit.
- Home Office segment generated the lowest sales and profit.
- Consumer segment represents the primary customer base.

Yearly Analysis

Insights:

- 2026 recorded the highest sales performance.
- 2024 recorded the lowest sales performance.

Loss Analysis

Further analysis revealed that:

- Furniture sales within the Consumer segment in the Central region generated negative profits.
- Profit margins were also negative in this combination.
- This indicates a potential issue related to pricing, discounts, or product mix.

7. Recommendations

Improve Furniture Profitability

Investigate the Furniture category to identify factors causing low profitability despite strong sales performance.

Analyze Central Region Performance

Review sales strategies and operational performance within the Central region to improve profit margins.

Investigate Loss-Making Segments

Focus on Furniture sales within the Consumer segment in the Central region where losses are occurring.

Optimize Product Mix

Evaluate product-level performance within Furniture and prioritize higher-margin products.

Leverage High-Performing Areas

Continue investing in Technology products and successful strategies used in the West region.

8. Conclusion

This project successfully transformed raw retail sales data into actionable business insights using Power Query and Power BI. Through data cleaning, KPI creation, dashboard development, and analysis, several performance trends and profitability concerns were identified. The project highlighted opportunities to improve business profitability, particularly within the Furniture category and Central region, while also identifying high-performing areas that can be leveraged for future growth.